24K—0559

BAZIL-UDDIN-KHAN

BSCS-1H

TASK 1).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

void Adddata(int * ptr, int Totalelements);

void Dma(int *& ptr, int Totalelements);
void Displayinfo(int * ptr, int Totalelements);

void Freememory(int * ptr, int Totalelements);



int main()
```

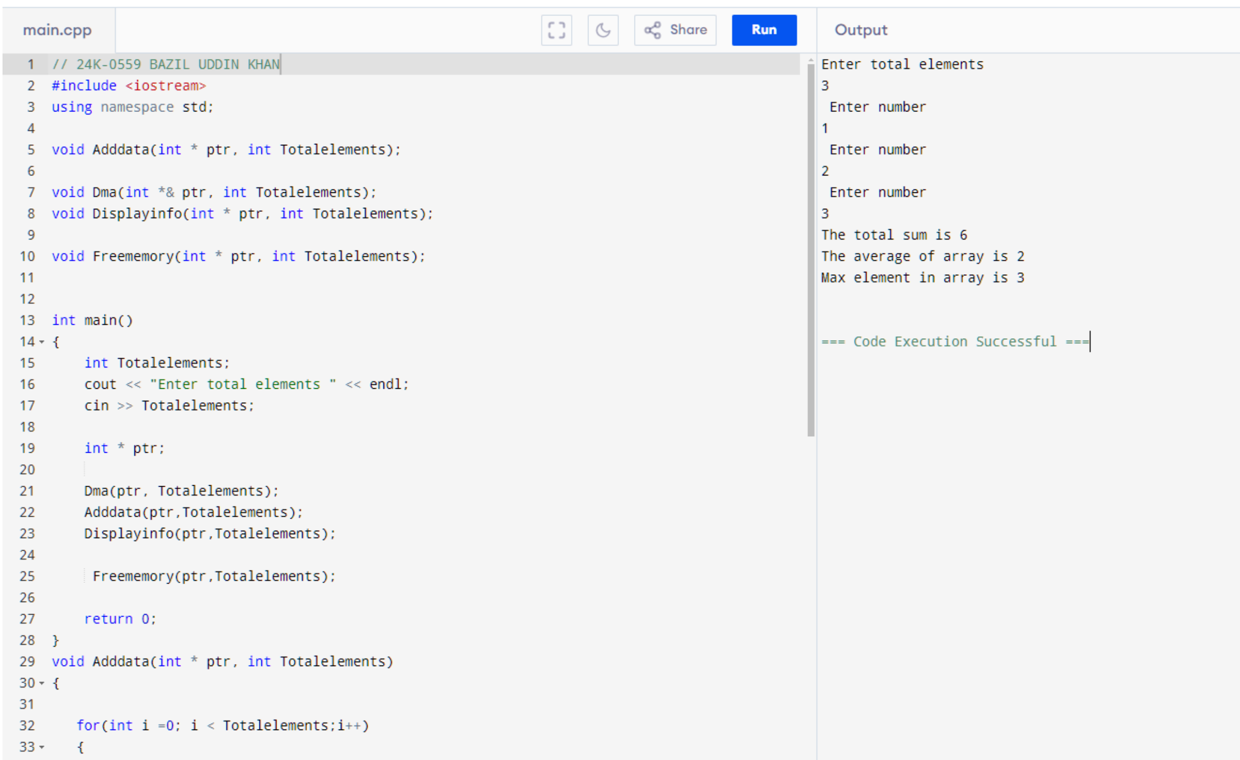
```
{
   int Totalelements;
   cout << "Enter total elements " << endl;
   cin >> Totalelements;

   int * ptr;

   Dma(ptr, Totalelements);
   Adddata(ptr,Totalelements);
   Displayinfo(ptr,Totalelements);

    Freememory(ptr,Totalelements);

   return 0;
}
void Adddata(int * ptr, int Totalelements)
{

  for(int i =0; i < Totalelements;i++)
```

```
  {
     int num;
     cout << " Enter number " << endl;
     cin >> num;
     ptr[i] = num;
  }
}

void Dma(int *& ptr, int Totalelements)
{
    ptr = new int[Totalelements];
}

void Displayinfo(int * ptr, int Totalelements)
{
   int Totalsum =0;
   for(int i =0; i < Totalelements;i++)
   {
      Totalsum = Totalsum + ptr[i];
```

```
    }
    cout << "The total sum is " << Totalsum << endl;
    float average = (Totalsum/Totalelements);
    cout << "The average of array is " << average << endl;
    int Max = ptr[0];
    for(int i =0; i < Totalelements;i++)
    {
        if(ptr[i] > Max)
        {
            Max = ptr[i];
        }
    }

    cout << "Max element in array is " << Max << endl;

}

void Freememory(int * ptr, int Totalelements)
{
```

```
    delete [] ptr;
}
```

OUTPUT:



TASK#2).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;
```

```
void Adddata(int ** ptrone, int ** ptrtwo, int
rowsone,int colsone,int rowstwo,int colstwo);

void Dma(int **& ptrone, int **& ptrtwo, int rowsone,
int colsone,int rowstwo, int colstwo);
void Addition(int ** ptrone,int **ptrtwo, int rowsone,
int colsone,int rowstwo, int colstwo);
void Subtraction(int ** ptrone,int ** ptrtwo, int
rowsone, int colsone,int rowstwo, int colstwo);
void Multiplication(int ** ptrone,int ** ptrtwo, int
rowsone, int colsone,int rowstwo, int colstwo);
void Freememory(int ** ptrone, int ** ptrtwo,int
rowsone, int rowstwo);


int main()
{
   int rowsone;
   cout << "Enter total rows for Array 1 " << endl;
```

```
   cin >> rowsone;
   int colsone;
   cout << "Enter total cols  for Array 1 " << endl;
   cin >> colsone;
    int rowstwo;
   cout << "Enter total rows for Array 2" << endl;
   cin >> rowstwo;
   int colstwo;
   cout << "Enter total cols  for Array 2 " << endl;
   cin >> colstwo;

   int ** ptrone;
   int ** ptrtwo;

Dma(ptrone,ptrtwo,rowsone,colsone,rowstwo,colstwo);

Adddata(ptrone,ptrtwo,rowsone,colsone,rowstwo,colst
wo);
   if(rowsone == rowstwo && colsone == colstwo &&
rowstwo == colstwo)
```

```
   {
Addition(ptrone,ptrtwo,rowsone,colsone,rowstwo,colst
wo);
      Subtraction(ptrone, ptrtwo,  rowsone,  colsone,
rowstwo,colstwo);

   }
   if(rowsone == colstwo)
   {
Multiplication(ptrone,ptrtwo,rowsone,colsone,rowstwo
,colstwo);
   }

   Freememory(ptrone,ptrtwo,rowsone,rowstwo);

   return 0;
}
```

```
void Adddata(int ** ptrone, int ** ptrtwo, int
rowsone,int colsone,int rowstwo,int colstwo)
{

  for(int i =0; i < rowsone;i++)
  {
     for(int j =0; j < colsone; j++)
     {
        int num;
       cout << " Enter number for array one " << endl;
       cin >> num;
       ptrone[i][j] = num;

     }

  }
  for(int i =0; i < rowstwo;i++)
  {
     for(int j =0; j < colstwo; j++)
```

```
      {
         int num;
        cout << " Enter number for array two " << endl;
        cin >> num;
        ptrtwo[i][j] = num;


      }


   }
}

void Dma(int **& ptrone, int **& ptrtwo, int rowsone,
int colsone,int rowstwo, int colstwo)


{
    ptrone = new int * [rowsone];
    for(int i =0; i < rowsone;i++)
    {
       ptrone[i] = new int[colsone];
```

```
    }

    ptrtwo = new int * [rowstwo];
    for(int u = 0; u < rowstwo;u++)
    {
        ptrtwo[u] = new int[colstwo];
    }
}
void Addition(int ** ptrone,int **ptrtwo, int rowsone,
int colsone,int rowstwo, int colstwo)
 {
    cout << "Addition answer is " << endl;

    int Sum[rowsone][colsone];
    for(int i =0; i < rowsone;i++)
     {
        for(int j =0; j < colsone;j++)
        {
            Sum[i][j] = ptrone[i][j] + ptrtwo[i][j];
```

```
      }
    }
    for(int i =0; i < rowsone;i++)
    {
      for(int j =0; j < colsone;j++)
      {
        cout << " " << Sum[i][j];
      }
      cout << endl;
    }


}
void Subtraction(int ** ptrone,int ** ptrtwo, int
rowsone, int colsone,int rowstwo, int colstwo)
{
  cout << "Subtraction answer is " << endl;


   int Subtract[rowsone][colsone];
```

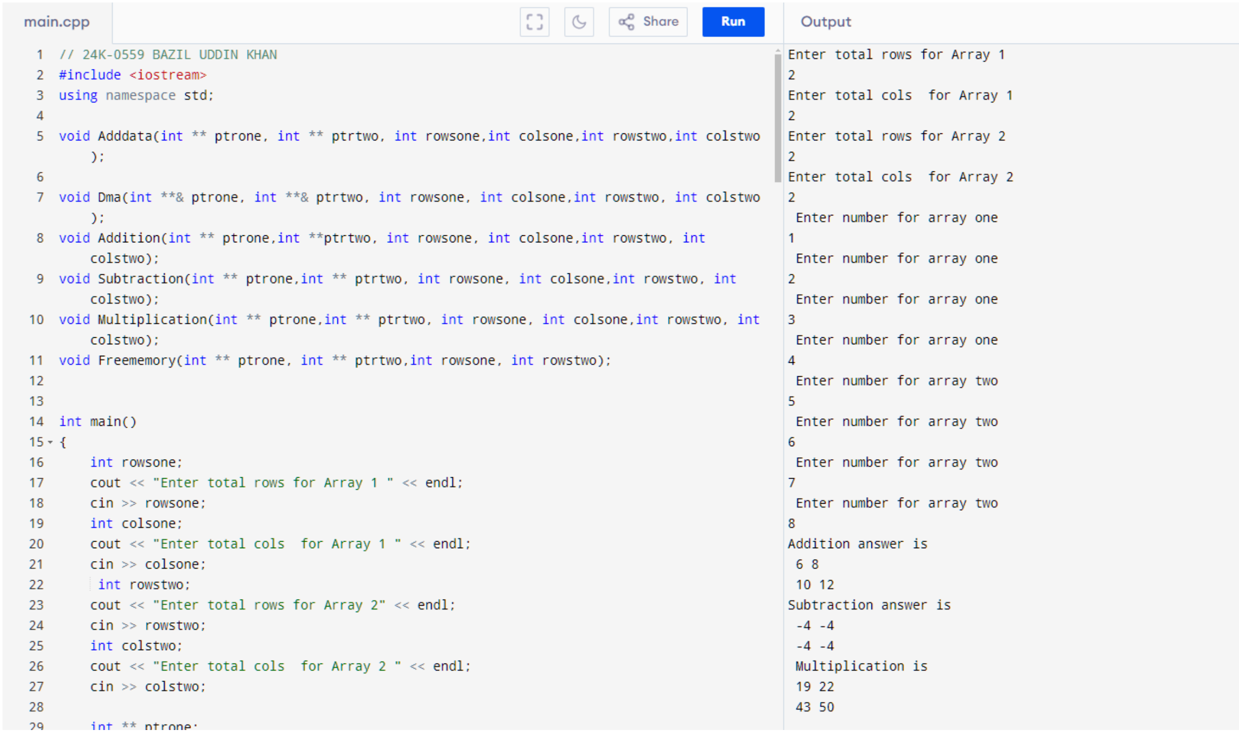
```
    for(int i =0; i < rowsone;i++)
    {
        for(int j =0; j < colsone;j++)
        {
            Subtract[i][j] = ptrone[i][j] - ptrtwo[i][j];
        }
    }
    for(int i =0; i < rowsone;i++)
    {
        for(int j =0; j < colsone;j++)
        {
           cout << " " << Subtract[i][j];
        }
        cout << endl;
    }

}
void Multiplication(int ** ptrone,int ** ptrtwo, int
rowsone, int colsone,int rowstwo, int colstwo)
```

```
{
   cout << " Multiplication is " << endl;
   int  Multiply[rowsone][colstwo];
   for(int i =0; i < rowsone;i++)
    {
      for(int j =0; j < colstwo;j++)
      {
        Multiply[i][j] = 0;
      }

    }

   for(int i =0; i < rowsone; i++)
   {
     for(int j = 0; j < colstwo; j++)
     {
       for(int k =0; k < colsone; k++)
       {
```

```
            Multiply[i][j] = Multiply[i][j] +  (ptrone[i][k] *
ptrtwo[k][j]);


         }




      }
   }
   for(int i =0; i < rowsone;i++)
    {
      for(int j =0; j < colstwo;j++)
      {
         cout << " " << Multiply[i][j];
      }
      cout << endl;
    }


}
```

```
void Freememory(int ** ptrone, int ** ptrtwo,int
rowsone, int rowstwo)
{
  for(int i =0; i < rowsone;i++)
  {
     delete ptrone[i];


  }
  delete [] ptrone;
}
```

OUTPUT:


main.cpp

Share Run

```
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

void Adddata(int ** ptrone, int ** ptrtwo, int rowsone,int colsone,int rowstwo,int colstwo
    );

void Dma(int **& ptrone, int **& ptrtwo, int rowsone, int colsone,int rowstwo, int colstwo
    );
void Addition(int ** ptrone,int **ptrtwo, int rowsone, int colsone,int rowstwo, int
    colstwo);
void Subtraction(int ** ptrone,int ** ptrtwo, int rowsone, int colsone,int rowstwo, int
    colstwo);
void Multiplication(int ** ptrone,int ** ptrtwo, int rowsone, int colsone,int rowstwo, int
    colstwo);
void Freememory(int ** ptrone, int ** ptrtwo,int rowsone, int rowstwo);


int main()
{
    int rowsone;
    cout << "Enter total rows for Array 1 " << endl;
    cin >> rowsone;
    int colsone;
    cout << "Enter total cols  for Array 1 " << endl;
    cin >> colsone;
     int rowstwo;
    cout << "Enter total rows for Array 2" << endl;
    cin >> rowstwo;
    int colstwo;
    cout << "Enter total cols  for Array 2 " << endl;
    cin >> colstwo;

    int ** ptrone;
```

Output

```
Enter total rows for Array 1
2
Enter total cols  for Array 1
2
Enter total rows for Array 2
2
Enter total cols  for Array 2
2
 Enter number for array one
1
 Enter number for array one
2
 Enter number for array one
3
 Enter number for array one
4
 Enter number for array two
5
 Enter number for array two
6
 Enter number for array two
7
 Enter number for array two
8
Addition answer is
 6 8
 10 12
Subtraction answer is
 -4 -4
 -4 -4
 Multiplication is
 19 22
 43 50
```

TASK#3).

CODE:

// 24K-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

struct Employee

{

```
    string Employeeid;
    string Name;
    string Department;
    double Salary;

};

void Addinfo(Employee * employee, int Totalemployees);
void Displayemployees(Employee * employee, int Totalemployees);
void Searchemployee(Employee * employee, int Totalemployees);
void Freememory(Employee * employee, int Totalemployees);

int main()
{

    int Totalemployees;
```

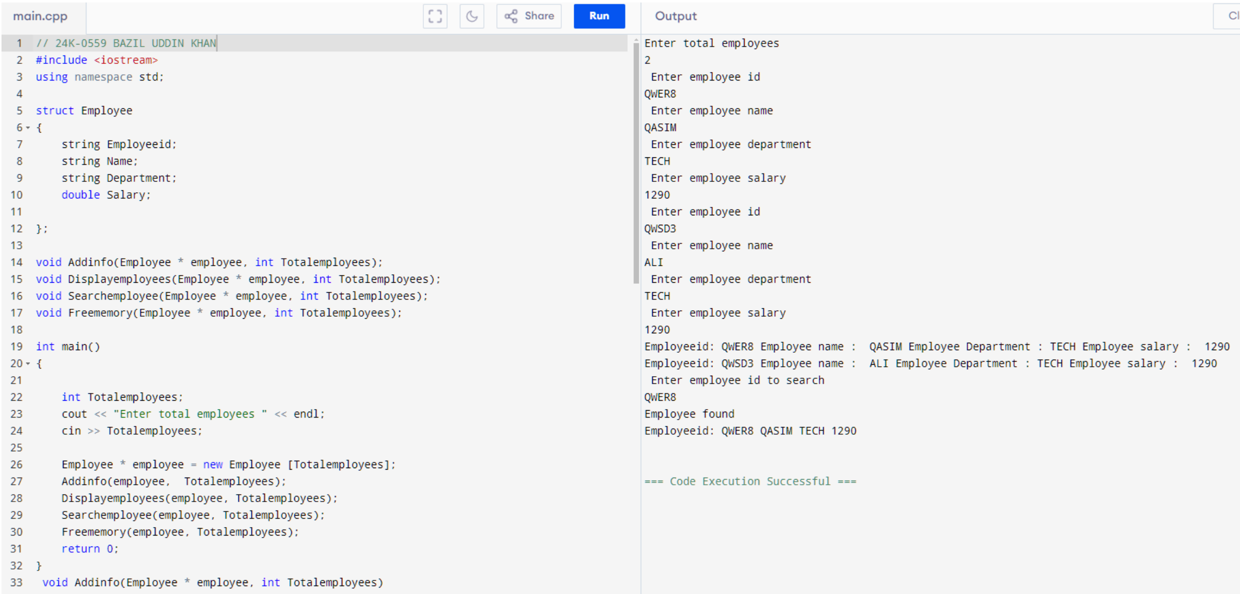
```
   cout << "Enter total employees " << endl;
   cin >> Totalemployees;


   Employee * employee = new Employee
[Totalemployees];
   Addinfo(employee,  Totalemployees);
   Displayemployees(employee, Totalemployees);
   Searchemployee(employee, Totalemployees);
   Freememory(employee, Totalemployees);
   return 0;
}
 void Addinfo(Employee * employee, int
Totalemployees)
 {
    for(int i =0; i < Totalemployees;i++)
     {
        string employeeid;
        cout << " Enter employee id " << endl;
        cin >> employeeid;
```

```
        string name;
        cout << " Enter employee name " << endl;
        cin >> name;
        string department;
        cout << " Enter employee department " << endl;
        cin >> department;
        double  salary;
        cout << " Enter employee salary  " << endl;
        cin >> salary;
        employee[i].Employeeid = employeeid;
        employee[i].Name = name;
        employee[i].Department = department;
        employee[i].Salary = salary;


    }
}

void Displayemployees(Employee * employee, int
Totalemployees)
```

```
{
    for(int i =0; i < Totalemployees;i++)
     {
        cout << "Employeeid: " << employee[i].Employeeid
<< " " << "Employee name : " <<
           " " <<  employee[i].Name << " " << "Employee
Department : " << employee[i].Department << " " <<
"Employee salary : " << " " <<  employee[i].Salary <<
endl;
     }


}
void Searchemployee(Employee * employee, int
Totalemployees)
{
    string EmployeeId;
    cout << " Enter employee id to search " << endl;
    cin >> EmployeeId;
    for(int i = 0; i < Totalemployees;i++)
     {
```

```
        if(employee[i].Employeeid == EmployeeId)
        {
            cout << "Employee found " << endl;
            cout << "Employeeid: " <<
employee[i].Employeeid << " " << employee[i].Name <<
" " << employee[i].Department << " " <<
employee[i].Salary << endl;

        }
    }

}
void Freememory(Employee * employee, int
Totalemployees)
{
    delete [] employee;
}
```

OUTPUT:



TASK#4).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

struct Student
{
```

```
    string Name;
    int Rollnumber ;
    int Marks[5];

};

void Addinfo(Student * student, int TotalStudents);
void Calculationanddisplay(Student * student, int
TTotalStudents,float * average );
void Freememory(Student * student, int
TotalStudents,float * average); // same

int main()
{

    int TotalStudents;
    cout << "Enter total students " << endl;
    cin >> TotalStudents;
    Student  * student  = new Student [TotalStudents];
```

```
   float * average = new float [TotalStudents];
   Addinfo(student,TotalStudents);
  // DisplayResults(student,TotalStudents);
   Calculationanddisplay(student, TotalStudents, average
);
   //(student, TotalStudents,average);
   Freememory(student, TotalStudents,average);
   return 0;
}
 void Addinfo(Student * student, int TotalStudents)
 {
    for(int i =0; i < TotalStudents;i++)
     {
        string name;
        cout << " Enter employee name  " << endl;
        cin >> name;
        int RollNumbeR;
        cout << " Enter employee roll number :  " << endl;
        cin >> RollNumbeR;
```

```
    for(int i =0; i <5; i++)
     {
       int Mark;
       cout << " Enter Student Mark : " << endl;
       cin >> Mark;
       student[i].Marks[i] = Mark;
     }

  }
}

void Calculationanddisplay(Student * student, int TotalStudents,float  * average)

{

  for(int i = 0; i < TotalStudents;i++)
   {
```

```
int Totalmarks = 0;
for(int j =0; j <5; j++)
 {
    Totalmarks = Totalmarks + student[i].Marks[i];

 }
 float Average = (Totalmarks/5);
 average[i] = Average;
 if(Average >= 90 )
  {
     cout << "Your Grade is A+"<<" for student " <<
        "  " << "Marks are :  " <<average[i] << " "  <<
i+1 << endl;
  }
  else if(Average >=80 && Average < 90)
   {
      cout << "Your grade is A " << "for student " <<
" " << "Marks are :  "<< average[i] << " " << i+1 << endl;
   }
```

```
else if(Average >= 70 && Average < 80)
{
    cout << "Your grade is B " <<"for student "  <<"
" << "Marks are " << average[i] <<
      " " << i+1 <<endl;
}

else if(Average >=60 && Average < 70)
{
    cout << "Your grade is C " << "for student " <<
"  " << "Marks are " << average[i] <<
      "  " << i+1 << endl;


}
else if(Average >=50 && Average <60)
{
    cout << "Your grade is D " << "for student "
<< " " << " Marks are " << average[i] << " " << i+1 <<
endl;
```

```
            }
            else
             {
                 cout <<"Your grade  is E " <<"for student " <<
" " << "Marks are "  << average[i] << " " << i+1 << endl;
             }
     }


 }
void Freememory(Student * student, int
TotalStudents,float * average)
 {
    delete [] student;
    delete [] average;
 }
```

OUTPUT:

main.cpp

```cpp
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

struct Student
{
    string Name;
    int Rollnumber ;
    int Marks[5];

};

void Addinfo(Student * student, int TotalStudents);
void Calculationanddisplay(Student * student, int TTotalStudents,float * average );
void Freememory(Student * student, int TotalStudents,float * average); // same

int main()
{

    int TotalStudents;
    cout << "Enter total students " << endl;
    cin >> TotalStudents;
    Student  * student  = new Student [TotalStudents];
    float * average = new float [TotalStudents];
    Addinfo(student,TotalStudents);
   // DisplayResults(student,TotalStudents);
    Calculationanddisplay(student, TotalStudents, average );
    //(student, TotalStudents,average);
    Freememory(student, TotalStudents,average);
    return 0;
}
 void Addinfo(Student * student, int TotalStudents)
 {
     for(int i =0; i < TotalStudents;i++)
```

Output

```
Enter total students
2
 Enter employee name
QASIM
 Enter employee roll number :
1234
 Enter Student Mark :
10
 Enter Student Mark :
45
 Enter Student Mark :
60
 Enter Student Mark :
30
 Enter Student Mark :
79
 Enter employee name
ALI
 Enter employee roll number :
1905
 Enter Student Mark :
18
 Enter Student Mark :
40
 Enter Student Mark :
39
 Enter Student Mark :
90
 Enter Student Mark :
55
Your grade  is E for student  Marks are 18 1
Your grade  is E for student  Marks are 40 2
```

TASK#5).

CODE:

// 24K-0559 BAZIL UDDIN KHAN

#include <iostream>

#include <string>

```
using namespace std;



void Dma(string *& ptr , string Stringone, string
Stringtwo);
void Concatatestring(string * ptr, string Stringone,string
Stringtwo);
void Comparestring(string * ptr,string Stringone, string
Stringtwo);
void Lengthstring(string * ptr, string Stringone, string
Stringtwo);
void Freememory(string * ptr);

int main()
{
   string Stringone;
   string Stringtwo;
   cout << "Enter string one " << endl;
   cin >> Stringone;
   cout << "Enter string two " << endl;
```

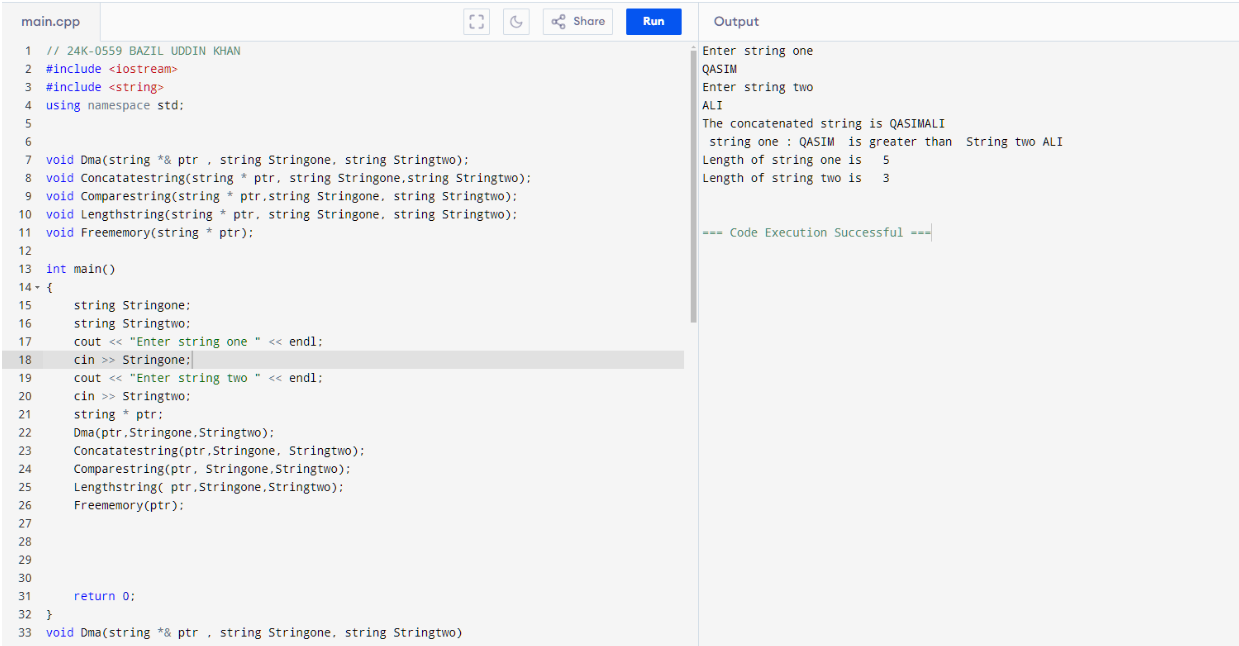
```
    cin >> Stringtwo;
    string * ptr;
    Dma(ptr,Stringone,Stringtwo);
    Concatatestring(ptr,Stringone, Stringtwo);
    Comparestring(ptr, Stringone,Stringtwo);
    Lengthstring( ptr,Stringone,Stringtwo);
    Freememory(ptr);




    return 0;
}
void Dma(string *& ptr , string Stringone, string
Stringtwo)
{
    ptr = new string[2];
    ptr[0] = Stringone;
    ptr[1] = Stringtwo;
```

```
}
void Concatatestring(string * ptr, string Stringone,string
Stringtwo)
{
   cout << "The concatenated string is " <<
(Stringone+Stringtwo) << endl;
}
void Comparestring(string * ptr,string Stringone, string
Stringtwo)
{
   if(Stringone == Stringtwo)
   {
      cout << "Both string one : " << Stringone << " " << "
and String two " << Stringtwo << " " << " is equal "
<<endl;
   }
  else  if(Stringone  > Stringtwo)
   {
```

```
        cout << " string one : " << Stringone << " " << " is
greater than  String two " << Stringtwo  <<endl;
    }
   else if(Stringone <  Stringtwo)
    {
        cout << " Stringtwo : " << Stringtwo << " " << " is
greater than  String one " << Stringone  <<endl;
    }



}
void Lengthstring(string * ptr, string Stringone, string
Stringtwo)
{

    int Lengthone = 0;
    int Lengthtwo = 0;
    for(int i =0; Stringone[i] != '\0' ;i++)
    {
```

```
        Lengthone++;
    }
    for(int i =0; Stringtwo[i] !='\0';i++)
    {
        Lengthtwo++;
    }

    cout << "Length of string one is "<< "  " << Lengthone
<< endl;
    cout << "Length of string two is " << "  " << Lengthtwo
<< endl;
}
void Freememory(string * ptr)
{
    delete [] ptr;
}
```

OUTPUT:



```
main.cpp

// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
#include <string>
using namespace std;


void Dma(string *& ptr , string Stringone, string Stringtwo);
void Concatatestring(string * ptr, string Stringone,string Stringtwo);
void Comparestring(string * ptr,string Stringone, string Stringtwo);
void Lengthstring(string * ptr, string Stringone, string Stringtwo);
void Freememory(string * ptr);

int main()
{
    string Stringone;
    string Stringtwo;
    cout << "Enter string one " << endl;
    cin >> Stringone;
    cout << "Enter string two " << endl;
    cin >> Stringtwo;
    string * ptr;
    Dma(ptr,Stringone,Stringtwo);
    Concatatestring(ptr,Stringone, Stringtwo);
    Comparestring(ptr, Stringone,Stringtwo);
    Lengthstring( ptr,Stringone,Stringtwo);
    Freememory(ptr);




    return 0;
}
void Dma(string *& ptr , string Stringone, string Stringtwo)
{
```

```
Output

Enter string one
QASIM
Enter string two
ALI
The concatenated string is QASIMALI
 string one : QASIM  is greater than  String two ALI
Length of string one is   5
Length of string two is   3


=== Code Execution Successful ===
```

TASK#6).

CODE:

// 24k-0559 BAZIL UDDIN KHAN

```
#include <iostream>
using namespace std;


struct Book
{
    string Bookid;
```

```
    string Title;
    string Author;
    string Isavailable;

};

void Addinfo(Book  * book, int Totalbooks);
void Displaybooks(Book * book, int Totalbooks);
void Searchbooks(Book * book, int Totalbooks);
void Freememory(Book* book, int Totalbooks);
void Removebook(Book *& book, int & Totalbooks);

int main()
{
    cout << "Welcome to library Management system " <<
endl;

    int Totalbooks;
    cout << "Enter total Books " << endl;
```

```
    cin >> Totalbooks;


    Book  * book = new Book [Totalbooks];
    Addinfo(book,Totalbooks);
    Displaybooks(book,Totalbooks);
    Searchbooks(book, Totalbooks);


    Removebook(book,Totalbooks);
    Displaybooks(book,Totalbooks);
    Freememory(book, Totalbooks);


    return 0;
}
 void Addinfo(Book * book, int Totalbooks)
{
     for(int i =0; i < Totalbooks;i++)
      {
         string bookid;
         cout << " Enter book  id " << endl;
```

```
        cin >> bookid;
        string title;
        cout << " Enter book title  " << endl;
        cin >> title;
        string author;
        cout << " Enter book author " << endl;
        cin >> author;
        string isAvailable;
        cout << " Enter in yes/no book status  " << endl;
        cin >> isAvailable;
        book[i].Bookid = bookid;
        book[i].Title = title;
        book[i].Author = author;
        book[i].Isavailable = isAvailable;


    }
}

void Displaybooks(Book * book, int Totalemployees)
```

```
{
    for(int i =0; i < Totalemployees;i++)
     {
        if(book[i].Isavailable == "yes")
        {
          cout << "Book id: " << book[i].Bookid << " " <<
"Book title : " << " " <<  book[i].Title << " " << "Book
Author is  "  << book[i].Author << " " << "book
availablity is : " << " " <<  book[i].Isavailable << endl;
        }
     }


}
void Searchbooks(Book * book, int Totalbooks)
{
    string bookId;
    cout << " Enter book id to search " << endl;
    cin >> bookId;
    for(int i = 0; i < Totalbooks;i++)
```

```
    {
        if(book[i].Bookid == bookId)
        {
            cout << "Book found " << endl;
            cout << "Book id: " << book[i].Bookid << " " <<
book[i].Title << " " << book[i].Author << " " <<
book[i].Isavailable << endl;


        }
    }


}
void Removebook(Book *& book, int  & Totalbooks)
{
   int index = -1;
   string bookid;
   cout << "Enter book id whose informationistobe
removed " << endl;
   cin >> bookid;
```

```
for(int i =0; i < Totalbooks;i++)
 {
    if(book[i].Bookid == bookid)
     {
        index = i;
     }
 }
 if(index == -1)
 {
    cout << "Not found " << endl;


 }
 else
  {
     Book * bok = new Book[Totalbooks - 1];
     int Index =0;
     for(int i = 0; i < Totalbooks;i++)
      {
         if(i != index)
```

```
            {
                bok[Index] = book[i];
                Index++;
            }


        }
        delete [] book;
        book = bok;

        Totalbooks --;
    }

}
void Freememory(Book * book, int Totalbooks)
 {
    delete [] book;
 }
```

OUTPUT:

main.cpp | Share | Run

```cpp
// 24k-0559  BAZIL UDDIN KHAN

#include <iostream>
using namespace std;

struct Book
{
    string Bookid;
    string Title;
    string Author;
    string Isavailable;

};

void Addinfo(Book  * book, int Totalbooks);
void Displaybooks(Book * book, int Totalbooks);
void Searchbooks(Book * book, int Totalbooks);
void Freememory(Book* book, int Totalbooks);
void Removebook(Book *& book, int & Totalbooks);

int main()
{
    cout << "Welcome to library Management system " << endl;

    int Totalbooks;
    cout << "Enter total Books " << endl;
    cin >> Totalbooks;

    Book  * book = new Book [Totalbooks];
    Addinfo(book,Totalbooks);
    Displaybooks(book,Totalbooks);
    Searchbooks(book, Totalbooks);

    Removebook(book,Totalbooks);
```

Output

```
Welcome to library Management system
Enter total Books
3
 Enter book  id
aqe
 Enter book title
fguo
 Enter book author
fgu
 Enter in yes/no book status
yes
 Enter book  id
vgju
 Enter book title
6rfr
 Enter book author
kcf
 Enter in yes/no book status
no
 Enter book  id
yfhc
 Enter book title
fuuf
 Enter book author
vk
 Enter in yes/no book status
yes
Book id: aqe Book title :  fguo Book Author is  fgu book availablity is :  yes
Book id: yfhc Book title :  fuuf Book Author is  vk book availablity is :  yes
 Enter book id to search
yfhc
Book found
Book id: yfhc fuuf vk yes
Enter book id whose informationistobe removed
```

main.cpp | Share | Run

```cpp
// 24k-0559  BAZIL UDDIN KHAN

#include <iostream>
using namespace std;

struct Book
{
    string Bookid;
    string Title;
    string Author;
    string Isavailable;

};

void Addinfo(Book  * book, int Totalbooks);
void Displaybooks(Book * book, int Totalbooks);
void Searchbooks(Book * book, int Totalbooks);
void Freememory(Book* book, int Totalbooks);
void Removebook(Book *& book, int & Totalbooks);

int main()
{
    cout << "Welcome to library Management system " << endl;

    int Totalbooks;
    cout << "Enter total Books " << endl;
    cin >> Totalbooks;

    Book  * book = new Book [Totalbooks];
    Addinfo(book,Totalbooks);
    Displaybooks(book,Totalbooks);
    Searchbooks(book, Totalbooks);

    Removebook(book,Totalbooks);
```


Output

```
aqe
 Enter book title
fguo
 Enter book author
fgu
 Enter in yes/no book status
yes
 Enter book  id
vgju
 Enter book title
6rfr
 Enter book author
kcf
 Enter in yes/no book status
no
 Enter book  id
yfhc
 Enter book title
fuuf
 Enter book author
vk
 Enter in yes/no book status
yes
Book id: aqe Book title :  fguo Book Author is  fgu book availablity is :  yes
Book id: yfhc Book title :  fuuf Book Author is  vk book availablity is :  yes
 Enter book id to search
yfhc
Book found
Book id: yfhc fuuf vk yes
Enter book id whose informationistobe removed
vgju
Book id: aqe Book title :  fguo Book Author is  fgu book availablity is :  yes
Book id: yfhc Book title :  fuuf Book Author is  vk book availablity is :  yes
```

TASK#7).

CODE:

```
// 24k-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

void Inputvalues(int * ptr, int Totalelements);
void Swapnum(int *x, int * y);
void Freememory(int *ptr);
void Reversearray(int *& ptr, int Totalelements);
void Display(int * ptr, int Totalelements);

int main()
{
   int x,y;
   cout << "Enter x " << endl;
   cin >>( x);
   cout << "Enter y " << endl;
   cin >> (y);
```

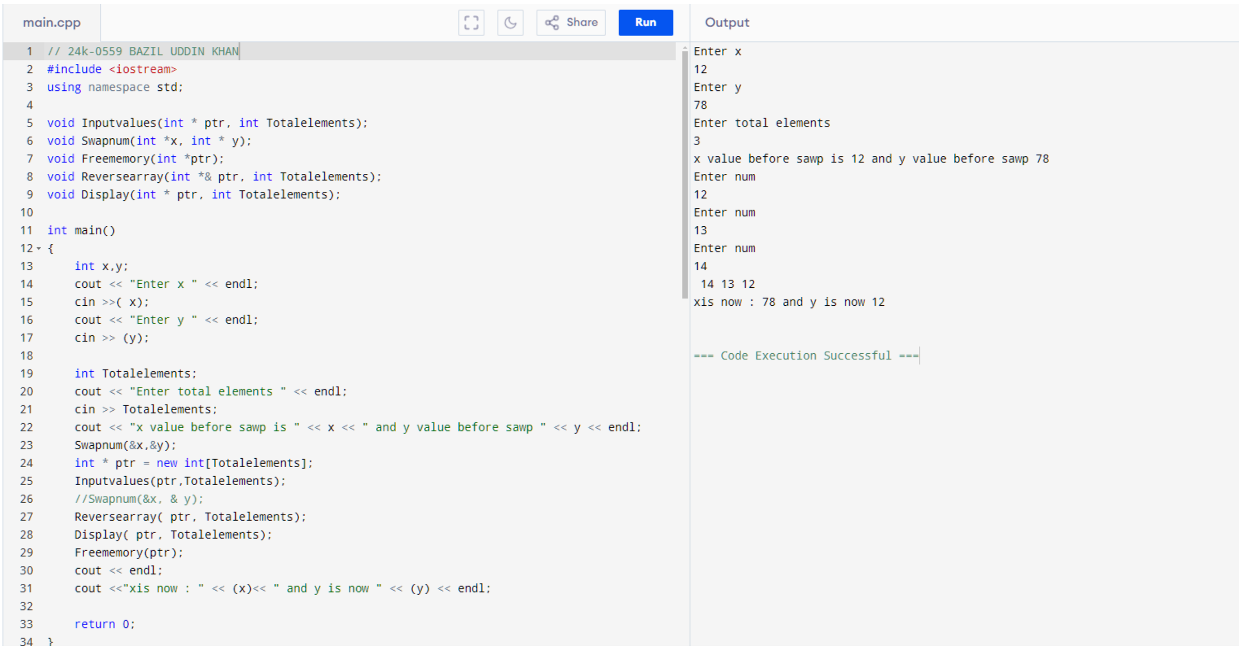
```
    int Totalelements;
    cout << "Enter total elements " << endl;
    cin >> Totalelements;
    cout << "x value before sawp is " << x << " and y value
before sawp " << y << endl;
    Swapnum(&x,&y);
    int * ptr = new int[Totalelements];
    Inputvalues(ptr,Totalelements);
    //Swapnum(&x, & y);
    Reversearray( ptr, Totalelements);
    Display( ptr, Totalelements);
    Freememory(ptr);
    cout << endl;
    cout <<"xis now : " << (x)<< " and y is now " << (y) <<
endl;

    return 0;
}
```

```
void Inputvalues(int * ptr, int Totalelements)
{
   for(int i =0; i < Totalelements;i++)
   {
      int num;
      cout << "Enter num "<<endl;
      cin >> num;
      ptr[i] = num;
   }

}

void Swapnum(int* x, int*  y)
{
   int Temp = *x;
   *x = *y;
   *y = Temp;
```

```
}

void Freememory(int *ptr)
{
    delete [] ptr;
}

void Reversearray(int *& ptr, int Totalelements)
{
    int * Array= new int [Totalelements];
   // for(int j =0; j < Totalelements;j++)
  //  {
     //  Array[j] = ptr[j];
   // }
   int Index = Totalelements-1;
    for(int i =0;i < Totalelements; i++)
    {
        Array[Index] = ptr[i];
        Index--;
```

```
    }

    for(int i = 0;i < Totalelements;i++)
    {
      (ptr[i]) = (Array[i]);
    }

    delete [] Array;



}
void Display(int * ptr, int Totalelements)
{
    for(int i = 0; i < Totalelements;i++)
    {
        // cout << "Revserse ordee num is " ;
        cout <<" " <<ptr[i];
    }
}
```

OUTPUT:



main.cpp

```cpp
// 24k-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

void Inputvalues(int * ptr, int Totalelements);
void Swapnum(int *x, int * y);
void Freememory(int *ptr);
void Reversearray(int *& ptr, int Totalelements);
void Display(int * ptr, int Totalelements);

int main()
{
    int x,y;
    cout << "Enter x " << endl;
    cin >>( x);
    cout << "Enter y " << endl;
    cin >> (y);

    int Totalelements;
    cout << "Enter total elements " << endl;
    cin >> Totalelements;
    cout << "x value before sawp is " << x << " and y value before sawp " << y << endl;
    Swapnum(&x,&y);
    int * ptr = new int[Totalelements];
    Inputvalues(ptr,Totalelements);
    //Swapnum(&x, & y);
    Reversearray( ptr, Totalelements);
    Display( ptr, Totalelements);
    Freememory(ptr);
    cout << endl;
    cout <<"xis now : " << (x)<< " and y is now " << (y) << endl;

    return 0;
}
```

Output

```
Enter x
12
Enter y
78
Enter total elements
3
x value before sawp is 12 and y value before sawp 78
Enter num
12
Enter num
13
Enter num
14
 14 13 12
xis now : 78 and y is now 12

=== Code Execution Successful ===
```

TASK#8).

CODE: // 24k-0559 BAZIL UDDIN KHAN

```cpp
#include <iostream>

using namespace std;


struct Product

{

   string Productid;
```

```
    string Name;
    int Quantity;
    double Price;

};

void Addinfo(Product * product, int Totalproducts);
void DisplayTotal(Product * product, int Totalproducts);
void Removeproduct(Product*& product, int
&Totalproducts);
void Updateproducts(Product *& product,int
Totalproducts);
void Freememory(Product * product, int Totalproducts);
void Display(Product * product, int Totalproducts);

int main()
{

    int Totalproducts;
```

```
    cout << "Enter total products " << endl;
    cin >> Totalproducts;

    Product * product = new Product [Totalproducts];
    Addinfo(product,Totalproducts);
    DisplayTotal(product,Totalproducts);
    Removeproduct(product,Totalproducts);
    Updateproducts(product,Totalproducts);
    Display( product, Totalproducts);
    Freememory(product,Totalproducts);
    return 0;
}
void Addinfo(Product * product, int Totalproducts)
 {
     for(int i =0; i < Totalproducts;i++)
      {
         string productid;
         cout << " Enter product id " << endl;
         cin >> productid;
```

```
        string name;
        cout << " Enter product name " << endl;
        cin >> name;
        int quantity;
        cout << " Enter priduct total quantity   " << endl;
        cin >> quantity;
        double price;
        cout << " Enter total products  of product " <<
endl;
        cin >> price;
        product[i].Productid = productid;
        product[i].Name = name;
        product[i].Quantity = quantity;
        product[i].Price = price;


    }
 }
 void DisplayTotal(Product * product, int Totalproducts)
 {
```

```
    double Total =0;
    for(int i =0; i < Totalproducts;i++)
     {
        Total = Total + (product[i].Quantity *
product[i].Price);
     }
     cout << " The total price of the prsent " << Total <<
endl;


 }


 void Removeproduct(Product*& product, int &
Totalproducts)
 {
   int index = -1;
   string productid;
   cout <<" Enter product id whose informationistobe
removed " << endl;
   cin >> productid;
   for(int i =0; i < Totalproducts;i++)
```

```
{
    if(product[i].Productid == productid)
     {
        index = i;
     }
}
if(index == -1)
{
    cout << "Not found " << endl;

}
else
 {
     Product * pro = new Product[Totalproducts - 1];
     int Index =0;
     for(int i = 0; i < Totalproducts;i++)
      {
         if(i != index)
         {
```

```
            pro[Index] = product[i];
            Index++;
          }



       }
      delete [] product;
      product = pro;
      Totalproducts --;


    }


 }

 void Updateproducts(Product *& product,int
Totalproducts)
 {
   string productid;
```

```
    cout << "Enter product id whose information to be
updated " << endl;
    cin >> productid;
    for(int i =0; i < Totalproducts;i++)
     {
        if(product[i].Productid == productid)
         {
            string Newproductid;
            cout << "Enter updated product id " << endl;
            cin >> Newproductid;
            string Newname;
            cout << "Enter updated name " << endl;
            cin >> Newname;
            int quantity;
            cout << "Enter updated Quantity " << endl;
            cin >> quantity;
            double price;
            cout << "Enter updated  price " << endl;
            cin >> price;
```

```
            product[i].Productid = Newproductid;
            product[i].Name = Newname;
            product[i].Quantity = quantity;
            product[i].Price = price;

            cout << endl;
            cout << "Successfully updated " << endl;

        }

    }

}
void Display(Product * product, int Totalproducts)
{
    for(int i =0; i < Totalproducts;i++)
    {
        cout << "Product id is :" << product[i].Productid <<
endl;
```

```
        cout << "Product name  is :" << product[i].Name <<
endl;
        cout << "Product quantity is :" <<
product[i].Quantity << endl;
        cout << "Product price is :" << product[i].Price <<
endl;



    }
 }
 void Freememory(Product * product, int Totalproducts)
 {
    delete [] product;
 }
```

OUTPUT:

main.cpp

```cpp
// 24k-0559  BAZIL UDDIN KHAN

#include <iostream>
using namespace std;

struct Product
{
    string Productid;
    string Name;
    int Quantity;
    double Price;

};

void Addinfo(Product * product, int Totalproducts);
void DisplayTotal(Product * product, int Totalproducts);
void Removeproduct(Product*& product, int &Totalproducts);
void Updateproducts(Product *& product,int Totalproducts);
void Freememory(Product * product, int Totalproducts);
void Display(Product * product, int Totalproducts);

int main()
{

    int Totalproducts;
    cout << "Enter total products " << endl;
    cin >> Totalproducts;

    Product * product = new Product [Totalproducts];
    Addinfo(product,Totalproducts);
    DisplayTotal(product,Totalproducts);
    Removeproduct(product,Totalproducts);
    Updateproducts(product,Totalproducts);
    Display( product, Totalproducts);
```

Output

```
Enter total products
3
 Enter product id
eseti7
 Enter product name
yfyfy
 Enter priduct total quantity
5
 Enter total products  of product
100
 Enter product id
ert0
 Enter product name
tyuio
 Enter priduct total quantity
6
 Enter total products  of product
300
 Enter product id
qwe
 Enter product name
qrty
 Enter priduct total quantity
10
 Enter total products  of product
100
 The total price of the prsent 3300
 Enter product id whose informationistobe removed
qwe
Enter product id whose information to be updated
ert0
Enter updated product id
ert1
Enter updated name
```

main.cpp

```cpp
// 24k-0559  BAZIL UDDIN KHAN

#include <iostream>
using namespace std;

struct Product
{
    string Productid;
    string Name;
    int Quantity;
    double Price;

};

void Addinfo(Product * product, int Totalproducts);
void DisplayTotal(Product * product, int Totalproducts);
void Removeproduct(Product*& product, int &Totalproducts);
void Updateproducts(Product *& product,int Totalproducts);
void Freememory(Product * product, int Totalproducts);
void Display(Product * product, int Totalproducts);

int main()
{

    int Totalproducts;
    cout << "Enter total products " << endl;
    cin >> Totalproducts;

    Product * product = new Product [Totalproducts];
    Addinfo(product,Totalproducts);
    DisplayTotal(product,Totalproducts);
    Removeproduct(product,Totalproducts);
    Updateproducts(product,Totalproducts);
    Display( product, Totalproducts);
```

Output

```
 Enter total products  of product
300
 Enter product id
qwe
 Enter product name
qrty
 Enter priduct total quantity
10
 Enter total products  of product
100
 The total price of the prsent 3300
 Enter product id whose informationistobe removed
qwe
Enter product id whose information to be updated
ert0
Enter updated product id
ert1
Enter updated name
ft
Enter updated Quantity
5
Enter updated  price
109

Successfully updated
Product id is :eseti7
Product name  is :yfyfy
Product quantity is :5
Product price is :100
Product id is :ert1
Product name  is :ft
Product quantity is :5
Product price is :109
```

TASK#9)

CODE:

```
// 24k-0559  BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

struct Student
{
   string Name;
   int Rollnumber ;
   int * Marks;


};

void Addinfo(Student * student, int TotalStudents,int Totalsubjarr[]);
void Displayhighest(Student * student, int TotalStudents,int * Total, int  Totalsubjarr[]);
void Freememory(Student * student, int TotalStudents,int * Total);
```

```
int main()
{

   int TotalStudents;
   cout << "Enter total students " << endl;
   cin >> TotalStudents;
   Student  * student  = new Student [TotalStudents];
   int * Total = new int[TotalStudents];
   int totalsubjarr[50] ;

   Addinfo(student,TotalStudents,totalsubjarr);
   Displayhighest(student, TotalStudents,Total
,totalsubjarr);

   Freememory(student, TotalStudents,Total);
   return 0;
 }
 void Addinfo(Student * student, int TotalStudents,int
Totalsubjarr[])
```

```
{
    int Totalsubj = 0;
    for(int i =0; i < TotalStudents;i++)
     {

        string name;
        cout << " Enter Student name  " << endl;
        cin >> name;
        student[i].Name = name;
        int RollNumbeR;
        cout << " Student roll number :  " << endl;
        cin >> RollNumbeR;
        student[i].Rollnumber = RollNumbeR;



        int Entertotalmarks;
        cout << " enter total subjectS marks " << " for
student  " << i+1 << endl;
        cin >> Entertotalmarks;
```

```
        student[i].Marks= new int [Entertotalmarks];
        for(int j =0; j <Entertotalmarks; j++)
         {
           int Mark;
           cout << " Enter Student Mark : " << endl;
           cin >> Mark;
           student[i].Marks[j] = Mark;


         }
         Totalsubjarr[Totalsubj] = Entertotalmarks;
         Totalsubj++;


     }
 }

void Displayhighest(Student * student, int
TotalStudents,int * Total,int  Totalsubjarr[])
 {
```

```
for(int i = 0; i < TotalStudents;i++)
 {
    int Totalmarks = 0;
    for(int j =0; j <Totalsubjarr[i]; j++)
     {
        Totalmarks = Totalmarks + student[i].Marks[j];


     }



     Total[i] = Totalmarks;


 }


 int Highestmarks = Total[0];
 int index = 0;
 for(int i = 0; i < TotalStudents;i++)
  {
     if(Highestmarks < Total[i])
```

```
        {
            Highestmarks = Total[i];
            index = i;
        }
    }

    cout << "The Student Name is " <<
student[index].Name << " " << " and the roll number is "
<< student[index].Rollnumber << " " << " and total
marks are " <<  Highestmarks << endl;



}
void Freememory(Student * student, int
TotalStudents,int * Total)
{
    delete [] student;
    delete [] Total;
}
```

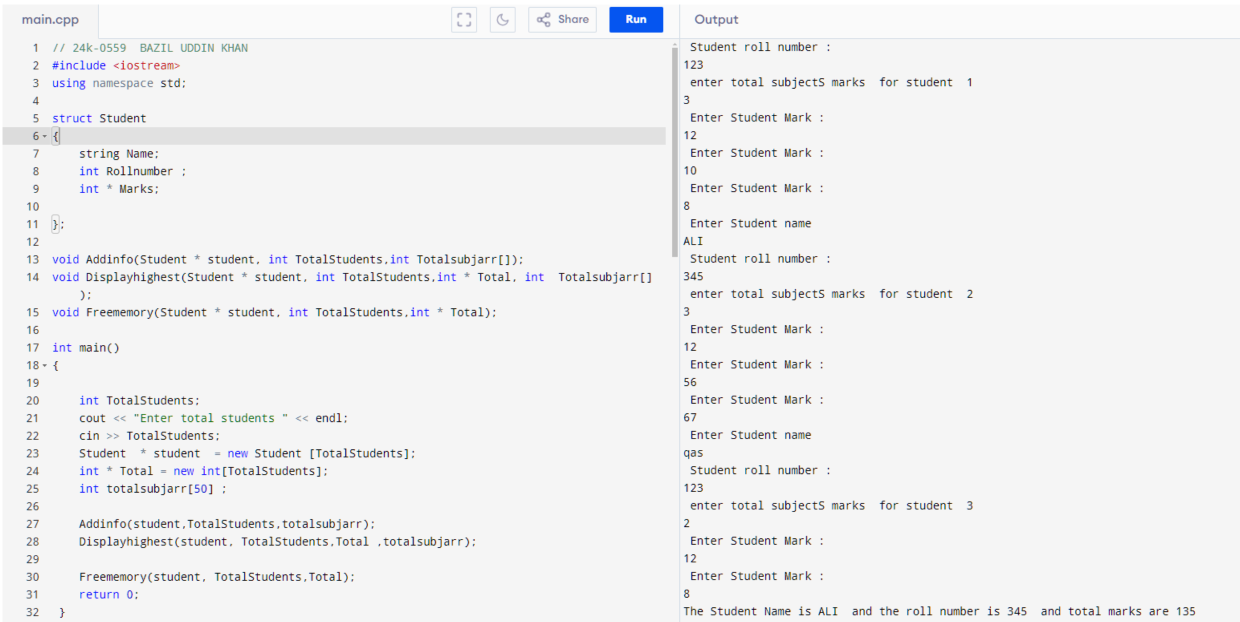
OUTPUT:

main.cpp

```cpp
// 24k-0559  BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

struct Student
{
    string Name;
    int Rollnumber ;
    int * Marks;

};

void Addinfo(Student * student, int TotalStudents,int Totalsubjarr[]);
void Displayhighest(Student * student, int TotalStudents,int * Total, int  Totalsubjarr[]
    );
void Freememory(Student * student, int TotalStudents,int * Total);

int main()
{

    int TotalStudents;
    cout << "Enter total students " << endl;
    cin >> TotalStudents;
    Student  * student  = new Student [TotalStudents];
    int * Total = new int[TotalStudents];
    int totalsubjarr[50] ;

    Addinfo(student,TotalStudents,totalsubjarr);
    Displayhighest(student, TotalStudents,Total ,totalsubjarr);

    Freememory(student, TotalStudents,Total);
    return 0;
 }
 void Addinfo(Student * student, int TotalStudents,int  Totalsubjarr[])
```

Output

```
Enter total students
3
 Enter Student name
QASIM
 Student roll number :
123
 enter total subjectS marks  for student  1
3
 Enter Student Mark :
12
 Enter Student Mark :
10
 Enter Student Mark :
8
 Enter Student name
ALI
 Student roll number :
345
 enter total subjectS marks  for student  2
3
 Enter Student Mark :
12
 Enter Student Mark :
56
 Enter Student Mark :
67
 Enter Student name
qas
 Student roll number :
123
 enter total subjectS marks  for student  3
2
 Enter Student Mark :
12
```



TASK#10).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;


void Adddata(int ** ptr, int rows,int cols);


void Dma(int **& ptr, int rows, int cols);
```

```
void Displayinfo(int ** ptr, int rows, int cols);

void Freememory(int ** ptr, int rows);


int main()
{
    int rows;
    cout << "Enter total rows " << endl;
    cin >> rows;
    int cols;
    cout << "Enter total cols " << endl;
    cin >> cols;

    int ** ptr;

    Dma(ptr, rows,cols);
    Adddata(ptr,rows,cols);
    Displayinfo(ptr,rows,cols);
```

```
    Freememory(ptr,rows);


    return 0;
}
void Adddata(int ** ptr, int rows, int cols)
{


   for(int i =0; i < rows;i++)
   {
      for(int j =0; j < cols; j++)
      {
          int num;
         cout << " Enter number " << endl;
         cin >> num;
         ptr[i][j] = num;


      }
```

```
  }
}

void Dma(int **& ptr, int rows, int cols)
{
     ptr = new int * [rows];
     for(int i =0; i < rows;i++)
     {
        ptr[i] = new int[cols];
     }
}

void Displayinfo(int ** ptr, int rows, int cols)
{
    cout << "Original Array is " << endl;
    for(int i = 0; i < rows;i++)
    {
       for(int j =0; j < cols;j++)
       {
```

```
        cout << " " << ptr[i][j];
    }
    cout << endl;
}

cout << "Transpose array is " << endl;
int Newarray[rows][cols];
int index =0;
for(int i =0; i < rows;i++)
{

    for(int j = 0; j < cols;j++)
    {
        Newarray[j][i] = ptr[i][j];
    }
}

for(int i = 0; i < rows;i++)
{
```

```
        for(int j =0; j < cols;j++)
        {
           cout << " " << Newarray[i][j];
        }
        cout << endl;
     }

}

void Freememory(int ** ptr, int rows)
{
   for(int i =0; i < rows;i++)
   {
      delete ptr[i];
   }
   delete [] ptr;
}
```

OUTPUT:

main.cpp Share Run

```cpp
// 24K-0559 BAZIL UDDIN KHAN
#include <iostream>
using namespace std;

void Adddata(int ** ptr, int rows,int cols);

void Dma(int **& ptr, int rows, int cols);
void Displayinfo(int ** ptr, int rows, int cols);

void Freememory(int ** ptr, int rows);


int main()
{
    int rows;
    cout << "Enter total rows " << endl;
    cin >> rows;
    int cols;
    cout << "Enter total cols " << endl;
    cin >> cols;

    int ** ptr;

    Dma(ptr, rows,cols);
    Adddata(ptr,rows,cols);
    Displayinfo(ptr,rows,cols);

    Freememory(ptr,rows);

    return 0;
}
void Adddata(int ** ptr, int rows, int cols)
{

```

Output

```
Enter total rows
2
Enter total cols
2
 Enter number
1
 Enter number
2
 Enter number
3
 Enter number
4
Original Array is
 1 2
 3 4
Transpose array is
 1 3
 2 4


=== Code Execution Successful ===
```